
Masked Diffusion Models as Associative Memories

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ABSTRACT

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1 Model

We study a teacher/student scenario for autoregressive models. We consider a dataset $\mathcal{D} = \{\mathbf{x}^\mu\}_{\mu=1}^M$ where sequences $\mathbf{x}^\mu \in \{-1, 1\}^L$ are generated by the stochastic process:

$$p(\mathbf{x} \mid W^*) = \prod_{i=1}^L \sigma \left(x_i \frac{1}{\sqrt{L}} \sum_{j<i} W_{ij}^* x_j \right) \quad (1)$$

given $W^* \in \mathbb{R}^{L \times L}$ fixed. Notice that we can take W^* to be lower triangular without the diagonal, and we choose $W_{ij}^* \sim N(0, 1)$ for $j < i$ and $W_{ij}^* = 0$ otherwise. The dataset is generated by sampling M sequences from the above distribution.

We then train a student model with matching architecture parametrised by W to generate sequences from the same distribution. $\hat{W}$ is obtained by minimizing the negative log-likelihood of the dataset:

$$\mathcal{L}(W) = -\frac{1}{M} \sum_{\mu=1}^M \log p(\mathbf{x}^\mu \mid W) = -\frac{1}{M} \sum_{\mu=1}^M \sum_{i=1}^L \log \sigma \left(x_i^\mu \frac{1}{\sqrt{L}} \sum_{j<i} W_{ij} x_j^\mu \right). \quad (2)$$

Due to factorization of the likelihood, minimisation happens component-wise on

$$\mathcal{L}_i(W) = -\frac{1}{M} \sum_{\mu=1}^M \log \sigma \left(x_i^\mu \frac{1}{\sqrt{L}} \sum_{j<i} W_{ij} x_j^\mu \right). \quad (3)$$

2 Replica Computation

The minimization problem can be reframed as a statistical mechanics problem. We optimise each row of $w_j = W_{ij}$ independently, and we can write the partition function for the i -th row as:

$$Z_i = \int d\mathbf{w} e^{-\beta \frac{\lambda}{2} \|\mathbf{w}\|^2 - M\beta \mathcal{L}_i(W)} = \int d\mathbf{w} e^{-\beta \frac{\lambda}{2} \|\mathbf{w}\|^2 + \beta \sum_{\mu=1}^M \log \sigma \left(x_i^\mu \frac{1}{\sqrt{L}} \sum_{j<i} w_j x_j^\mu \right)} \quad (4)$$

The replicated partition function is then:

$$\begin{aligned}
\mathbb{E}_{X, W^*}[Z_i^n] &= \mathbb{E}_{X, W^*} p \int \prod_a d\mathbf{w}^a e^{-\beta \frac{\lambda}{2} \sum_a \|\mathbf{w}_a\|^2 + \beta \sum_{\mu, a} \log \sigma\left(x_i^\mu \frac{1}{\sqrt{L}} \sum_{j < i} w_j^a x_j^\mu\right)} \\
&= \int \prod_a d\mathbf{w}^a e^{-\beta \frac{\lambda}{2} \sum_a \|\mathbf{w}_a\|^2} \mathbb{E}_{W^*} \prod_{\mu=1}^M \left[\mathbb{E}_{\mathbf{x}^\mu \mid W^*} e^{\beta \sum_a \log \sigma\left(x_i^\mu \frac{1}{\sqrt{L}} \sum_{j < i} w_j^a x_j^\mu\right)} \right] \\
&= \int \prod_a d\mathbf{w}^a e^{-\beta \frac{\lambda}{2} \sum_a \|\mathbf{w}_a\|^2} \int \prod_{ij} Dw_{ij}^* \left[\mathbb{E}_{\mathbf{x} \mid W^*} e^{\beta \sum_a \log \sigma\left(x_i \frac{1}{\sqrt{L}} \sum_{j < i} w_j^a x_j\right)} \right]^M.
\end{aligned} \tag{5}$$

Consider the generic sequence $\mathbf{x}$ generated by the teacher. For token i , we have

$$p(x_i) = \sigma(x_i h_i) \tag{6}$$

where $h_i = \frac{1}{\sqrt{L}} \sum_{j < i} W_{ij}^* x_j$. Since elements of different rows of W^* are independent, we can write the expectation over $\mathbf{x}$ as a product of expectations over each token:

Bibliography

